

HEMANTH THULASIRAMAN

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Education

University of Maryland

M.S in Data Science

Coursework: Probability and Statistics, Big Data Systems, Algorithms for Data Science

Aug 2025 – May 2027 (Expected)

Maryland, United States

Vellore Institute of Technology

B.Tech in Computer Science and Engineering

Coursework: NLP, Artificial Intelligence, Data Science, Database Systems, Computer Vision

Sep 2021 – May 2025

TamilNadu, India

Projects

CodeSense – Agentic Code Review System | *LangGraph, GPT-4o, Flake8, FastAPI, Supabase, Render* **Feb 2026**

- Built an event-driven code review agent triggered by GitHub webhooks, running Flake8 analysis as structured GPT-4o context - reducing review latency from **2 hours to under 60 seconds** at **< \$0.10/review**.
- Architected a **5-tool agent pipeline** where static analysis outputs ground LLM reasoning to specific line numbers and violation codes, deployed on Render with **LangSmith tracing** across all agent nodes.
- Implemented async **Human-in-the-Loop** approval via Slack with PostgreSQL state persistence and conditional escalation for HIGH severity Bandit findings before any review posts to GitHub.

FinSight – Agentic Financial Research System | *LangGraph, pgvector, LangSmith, Neon, Railway* **Dec 2025**

- Built an autonomous financial research agent using LangGraph **ReAct** loop and GPT-4o that compresses analyst research from hours to **90 seconds** - with self-healing retry logic when tools return low-coverage data.
- Architected a production retrieval stack with pgvector semantic memory, PostgreSQL persistence, and **Cloudflare R2** storage - enabling cross-run research grounding and delta reporting against prior briefs.
- Implemented two human-in-the-loop checkpoints, LangSmith token cost tracking, and variable task depth - achieving **100% task completion**, sub-10-second latency, and **\$0.007** cost per run across automated evaluation.

SupportMind – Customer Support Automation | *PyTorch, HuggingFace, PEFT, TRL, MLflow, Docker* **Oct 2025**

- Fine-tuned Mistral-7B using **LoRA** and 4-bit quantization for intent classification across 27 support categories, achieving **0.947** macro F1 on a held-out test set.
- Built an end-to-end ML pipeline for **data validation, stratified splitting, training, evaluation, and experiment tracking** with **MLflow** and Dockerized reproducibility across environments.
- Deployed the model as a **FastAPI** service with input validation and confidence monitoring, ensuring consistent behavior between training and production by sharing prompt formatting logic end-to-end.

Experience

OnMobile Global Limited

Data Science Intern, Business Intelligence Team

Jan 2025 – Jul 2025

Bengaluru, India

- Engineered and scaled an **LSTM time-series forecasting engine** for 40+ operators, achieving a **MAPE <10%** in predicting revenue and break-even points to drive operator-level profitability analysis.
- Deployed **production dashboards** integrating churn/retention KPIs with predictive forecasts; adopted by leadership for daily strategic spending and resource reallocation.
- Automated financial and marketing analysis workflows in **BigQuery**, reducing manual reporting time by **4 hours/week** and eliminating Excel-based data bottlenecks.

ELPIS IT Solutions Pvt. Ltd

Machine Learning Intern

Aug 2023 – Oct 2023

Bengaluru, India

- Architected an end-to-end evaluation framework that automated data preprocessing and parallel training for 5+ time-series models, including **ARIMA, Prophet, and LSTM**.
- Engineered a systematic ranking engine using **MAPE, RMSE, and MAE** metrics to eliminate manual selection bottlenecks and ensure reproducible forecasting performance.

Technical Skills

Languages: Python, SQL, Java, C/C++

AI Engineering: Claude Code, LangGraph, LangChain, CrewAI, RAG, Tool Calling, LLM Evaluation

ML Frameworks: Scikit-learn, XGBoost, PyTorch, TensorFlow, Hugging Face, SentenceTransformers

Backend & APIs: FastAPI, Flask, REST APIs, Pydantic, Uvicorn, n8n

Data & Infrastructure: PostgreSQL, pgvector, Redis, BigQuery, Neon, Supabase, Cloudflare R2, Docker, AWS, CI/CD, Git, GitHub, Railway, Render